Cours MALG & MOVEX

Vérification d'une annotation

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Plan

1. Correction
2. Règles appliquées sur les séquents

Sommaire

1. Correction

2. Règles appliquées sur les séquents

Current Summary

1. Correction

2. Règles appliquées sur les séquents

Ecriture du contrat

$$\ell_1 : x = 3 \ \wedge \ y = z{+}x \ \wedge z = 2{\cdot}x$$
$$y := z{+}x$$
$$\ell_2 : x = 3 \ \wedge \ y = x{+}6$$

On définit un contrat comme suit :

variables x, y, z
requires $x0 = 3 \wedge y0 = z0{+}x0 \wedge z0 = 2.x0$
ensures $x_f = 3 \wedge y_f = x_f{+}6$

begin
$\ell_1 : x = 3 \ \wedge \ y = z{+}x \ \wedge z = 2{\cdot}x$
$y := z{+}x$
$\ell_2 : x = 3 \ \wedge \ y = x{+}6$
end

Conditions de vérification du contrat

On pose les assertions suivantes à partir de l'annotation :

- $pre(x_0, y_0, z_0) \stackrel{def}{=} x0 = 3 \wedge y0 = z0{+}x0 \wedge z0 = 2.x0$
- $prepost(x_0, y_0, z_0, x, y, z) \stackrel{def}{=} x = 3 \ \wedge \ y = x{+}6$
- $Q_1(x_0, y_0, z_0, x, y, z) \stackrel{def}{=} x = 3 \ \wedge \ y = z{+}x \ \wedge z = 2{\cdot}x$
- $Q_2(x_0, y_0, z_0, x, y, z) \stackrel{def}{=} x = 3 \ \wedge \ y = x{+}6$

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On établit les trois cpnditions pour valider le contrat :

- (init) $pre(x_0, y_0, z_0) \wedge (x, y, z) = (x_0, y_0, z_0) \Rightarrow Q_1(x_0, y_0, z_0, x, y, z)$
- (concl) $pre(v_0) \wedge Q_2(x_0, y_0, z_0, x, y, z) \Rightarrow prepost(x_0, y_0, z_0, x, y, z)$
- (induct)
  $pre(x_0, y_0, z_0) \wedge Q_1(x_0, y_0, z_0, x, y, z) \wedge TRUE \wedge (x', y', z') =$
  $(x, z{+}x, z) \Rightarrow Q_2(x_0, y_0, z_0, x', y', z')$

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  - $x = 3$ est une hypothèse à gauche. Le séquent est valide.

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  - Réflexivité de l'égalité.

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